

```

[root@desktop1 ~]# journalctl -o verbose
-- Logs begin at Sun 2018-07-15 09:55:13 EDT, end at Sun 2018-07-15 10:15:14 EDT. --
Sun 2018-07-15 09:55:13.642743 EDT [s=33a33976eac1457ba8ed1257c767c504;i=1;b=23b3ea7e49cf4a
_PRIORITY=6
_TRANSPORT=driver
_MESSAGE=Runtime journal is using 8.0M (max 92.0M, leaving 138.0M of free 912.4M, curren
_MESSAGE_ID=ec387f577b844b8fa948f33cad9a75e6
_PID=89
_UID=0
_GID=0
_COMM=systemd-journal
_EXE=/usr/lib/systemd/systemd-journald
_CMDLINE=/usr/lib/systemd/systemd-journald
_CAP_EFFECTIVE=4402800cf
_SYSTEMD_CGROUP=/system.slice/systemd-journald.service
_SYSTEMD_UNIT=systemd-journald.service
_SYSTEMD_SLICE=system.slice
_SELINUX_CONTEXT=kernel
_BOOT_ID=23b3ea7e49cf4a07bd2c06f2a2c2012c
_MACHINE_ID=924b36755e244579ab08c33ba4f42732
_HOSTNAME=localhost
Sun 2018-07-15 09:55:13.688707 EDT [s=33a33976eac1457ba8ed1257c767c504;i=2;b=23b3ea7e49cf4a
_PRIORITY=6
_TRANSPORT=driver
_MESSAGE=Runtime journal is using 8.0M (max 92.0M, leaving 138.0M of free 912.4M, curren
_MESSAGE_ID=ec387f577b844b8fa948f33cad9a75e6
_PID=89
_UID=0

```

Figure 5: Verbose output

```
# journalctl --since today.
```

To output the journal entries from June 10, 2018 20:30:00 to June 13, 2018 12:00:00, use the following command:

```
# journalctl --since "2018-06-10 20:30:00" --until "2018-06-13 12:00:00."
```

In addition to the visible content of the journal, there are fields attached to log entries that can only be seen when verbose output is turned on. All of the displayed extra fields can be used to filter the output of a journal query. This is useful to reduce the output of complex searches for certain events in the journal.

Among the more useful options to search for lines relevant to a particular process or event are:

- **_COMM:** The name of the command
- **_EXE:** The path to the executable for the process
- **_PID:** The PID of the process
- **_UID:** The UID of the process
- **_SYSTEMD_UNIT:** The system unit that started the process

More than one of these can be combined. For example, the following query shows all journal entries related to processes started by the system unit file *sshd.service*, which also have the PID 1182:

```
#journalctl _SYSTEMD_UNIT=sshd.service _PID=1182
```

Note: For a list of commonly used journal fields, consult the man page of *systemd-journal-fields(7)* using the *man* command.

Preserving the *systemd* journal (storing the system journal permanently)

By default, the *systemd* journal is kept in */run/log/journal*, which

means it is cleared when the system reboots. The journal is a new mechanism in RHEL 7 and for most installations, a detailed journal that starts with the last boot is sufficient.

If the directory */var/log/journal* exists, the journal will log to that directory instead. The advantage of this is the historic data will be available immediately at boot. However, even with a persistent journal, not all dates will be kept forever. The journal has a built-in log rotation mechanism that will get triggered on a monthly basis. In addition, by default, the journal will not be allowed to become larger than 10 per cent of the file system it is on, or leave less than 15 per cent of the file system free. These values can be tuned in */etc/systemd/journal.conf* and the current limits on the size of the journal are logged when the *systemd-journald* process starts, as can be seen in the output of the *journalctl* command.

The *systemd* journal can be made persistent by creating the directory */var/log/journal* as the user root, with the following command:

```
# mkdir /var/log/journal
```

Ensure that the */var/log/journal* directory is owned by the root user and *systemd-journal*, and has the permissions 2755.

```
#chown root:systemd-journal /var/log/journal.
#chmod 2755 /var/log/journal
```

A reboot of the system on sending the special signal *USR1* as the root user to the *systemd-journald* process is required.

```
# killall -USR1 systemd-journald.
```

Since the *systemd* journal is now persistent across reboots, *journalctl -b* can reduce the output by only showing the log messages since the last boot of the system.

```
# journalctl -b
```

Note: When debugging a system crash with a persistent journal, it is usually required to limit the journal query to the reboot before the crash happened. The *-b* option can be accompanied by a negative number indicating how many prior system boots the output should be limited to. For example, *Journalctl -b -1* limits the output to the previous boot.

END

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